POLITECNICO MILANO 1863 Academic year 2023/2024	Course RF SYSTEMS	**Student ID or Persona code** _____
	Student Signature _____	**Student Name, Surname** _____

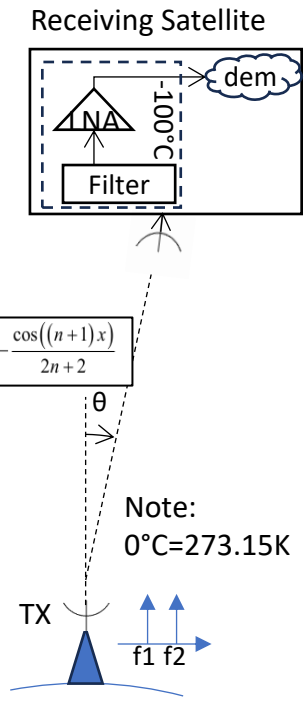
13 February 2024 Exam (27 pts available; ≥15 needed)

Please answer the following questions/problems, providing a brief explanation in English language of the steps/computations involved. Please specify units for all numeric results requiring them. Allowed support material: textbooks, notes, scientific calculator, laptop/tablet (offline). Write with a pen.

Exercise 1 (10 points)

An antenna on Earth radiates with directivity function $f(\theta,\phi)=\cos(200\cdot\theta)$ for θ in [0; π/400] (θ in radians) and 0 elsewhere, not depending on ϕ, with the θ=0 axis fixed toward sky (zenith). It is fed by a two-tone signal centered around 26GHz with spacing between the tones of 200MHz. The total average power provided to the antenna, perfectly matched, is 35dBm. The antenna efficiency is η=0.75.

The receiver is a low-Earth orbit satellite at *L*=1500km from the Earth surface, equipped with a 50dBi antenna always kept perfectly aligned toward the Earth transmitter, connected to a filter (A_f=1dB loss between 25 and 27GHz) and then to an LNA (G_{LNA}=28dB, $IIP3_{LNA}$=-65dBm) (Note: IIP3 is the input 3rd-order interc. Point; assume pure 3rd order non-linearity).



$$\int \sin(x)\cos(nx)\,dx = \frac{\cos((n-1)x)}{2n-2} - \frac{\cos((n+1)x)}{2n+2}$$

a) What is the transmitting antenna gain at 26GHz, in dBi?

b) In condition a), what is the received average power of the two-tone signal at the receiving antenna port when the satellite is exactly above the transmitter (at its zenith)?

c) Assuming instead that the received power of the 2-tone signal at the antenna port is PRx=-74.5dBm (different from what has been computed in b): Neglecting compression of the main tones and assuming pure third-order non-linearity, what is the equivalent SNIR ($P_{2ToneSignal}/(P_{ThermNoise}+P_{intermod})$) at the output of the LNA? Use the equivalent system noise of the receiver Tsys=120°C (Celsius, at the antenna port), measured on a signal bandwidth of 250MHz.

d) If the filter and LNA are cooled as in the figure and the equivalent noise temperature of the demodulator is 5500K (at its input) and the antenna picks up a noise temp. Ta=30K, what should the NF of the LNA be (in dB) to have Tsys=120°C? Remember that NF remains defined as $NF_{LNA}=1+T_{eqLNA}/290K$

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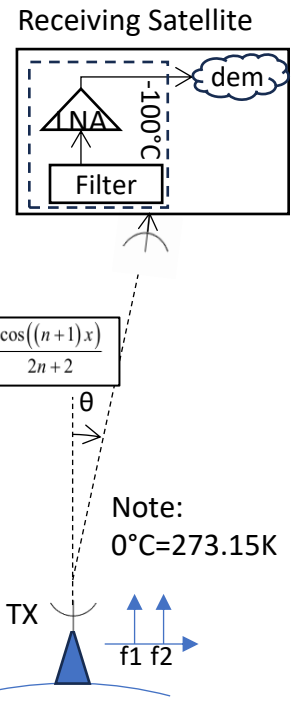
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SOLUTION

a) Tx antenna directivity, exploiting separability in ϕ and θ: $D_{Tx} = \frac{4\pi}{\int_0^{2\pi} d\phi \int_0^{\pi/400} \cos(200\cdot\theta)\sin(\theta)d\theta} = \frac{2}{\frac{\cos(199\,\theta)}{398} - \frac{\cos(201\theta)}{402}} = 140160\ (= 51.5dBi)$. The gain is GTx=DTx η=105120 (=50.2dBi)

b) Friis equation of link budget, with perfect reciprocal alignment, in log scale: with FSL=20Log(4 π L/λ)=184.3dB, PRx=PTx+GTx-FSL+GRx =-49.1dBm, where PTx=35dBm (average power of the 2-tone transmitted signal).

c) Using instead PRx=-74.5dBm, the main tones (at 25.9GHz and 26.1GHz) at LNA out have avg. power $P_{\omega1}=P_{\omega2}$=-74.5dBm-1dB-3dB+28dB=-50.5dBm each. The two intermodulation products (at 2f1-f2=25.7GHz and 2f2-f1=26.3GHz) have power $P_{2\omega2-\omega1}=P_{2\omega1-\omega2}=3P_{\omega1}-2IP3_{LNA}=3P_{\omega1}-2(IIP3_{LNA}+G_{LNA})$=-77.5dBm each.

The Equivalent thermal noise at the antenna port is Pneq=kTsysB=1.36pW (=-88.7dBm). Thus the equivalent noise at the LNA output is Pn_{LNAout}=Pneq-1dB+G_{LNA}=-61.7dBm. The combined power of noise and intermodulation products is $P_{n\&int}=10^{-61.7/10}$mW+ $2\cdot10^{-77.5/10}$mW=0.711nW (=-61.5dBm). The equivalent SNIR is SNIR=-50.5dBm+3dB-(-61.5dBm)=14dB. The "+3dB" is needed to include the avg power of the 2-tone main signal.

d) We write the expression of the equivalent system temperature: 393K=Tsys=Ta+Tf+TeqLNA/gf+Tdem/gf/gLNA
With gf=$10^{-1/10}$=0.79; Tf=173K(1/gf-1)=46K; gLNA=$10^{28/10}$=631. Thus T_{eqLNA}/gf=Tsys-Ta-Tf-Tdem/gf/gLNA=306K
Therefore: T_{eqLNA}=244.5K→ $NF_{LNA}=1+T_{eqLNA}/290K$=1.84 (=2.65dB)

ex 1 | h 16:09 | → | 34 MINUTES |

$f(\vartheta,\varphi) = \cos 200\vartheta \quad \vartheta \in [0, \frac{\pi}{400}] \ \forall\varphi$

$f_0 = 26\,GHz \quad \Delta f = 200\,MHz$

$P_{TX} = 35\,dBm \quad \eta = 0.75$

a)

$$D_{Tx} = \frac{4\pi}{\int_0^{2\pi} d\phi \int_0^{\pi/400} \cos(200\cdot\theta)\sin(\theta)d\theta} =$$

$$\int \sin(x)\cos(nx)\,dx = \frac{\cos((n-1)x)}{2n-2} - \frac{\cos((n+1)x)}{2n+2}$$

$$\frac{2}{\frac{\cos(199\,\theta)}{398} - \frac{\cos(201\theta)}{402}} = 140160\ (= 51.5dBi)$$

GTx=DTx η=105120 (=50.2dBi)

PTx=35dBm (average power of the 2-tone transmitted signal)

b) $P_{RX} = P_{TX} + G_{TX} - FSL + G_{RX} = -49.1\,dBm$

$FSL = 20\log\frac{4\pi d}{\lambda} = 184.3\,dB$

Using instead PRx=-74.5dBm,

c) SPECTRUM at LNA OUTPUT? (LNA G = 28 dB, IIP3 = -65 dBm)

$P_{@LNA\ INPUT} = -74.5\,dBm - 1\,dB = -75.5\,dBm$

(the two tones are at 25.9 GHz and 26.1 GHz, thus both in the PASSBAND where the filter loses only 1dB)

AT INPUT of LNA: $P'^{IN}_{tot} = P'^{IN}_{\omega_1} = -75.5\,dBm - 3\,dB = -78.5\,dBm$

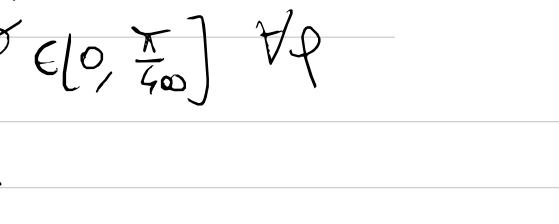
AT OUTPUT of LNA: $P_{\omega_1} = P_{\omega_2} = P'^{IN}_{\omega_1} + G_{LNA} = -50.5\,dBm$

INTERMOD TONES AT OUT: $P_{2\omega_2-\omega_1} = P_{2\omega_1-\omega_2} = 3P_{\omega_1} - 2IP_3$ (OUTPUT IP3)

$IP_3 = IIP_3 + G_{LNA} = -65\,dBm + 28\,dB = -37\,dBm$

$\Rightarrow P_{2\omega_2-\omega_1} = P_{2\omega_1-\omega_2} = -77.5\,dBm$

SPECTRUM AT OUTPUT



$2f_2 - f_1 = 26.3\,GHz$

$2f_1 - f_2 = 25.7\,GHz$

$\left(\Rightarrow C/I = \frac{P^{LIN\ SCALE}_{TWO\text{-}TONES\ SIGNAL\ AT\ OUT}}{P^{LIN\ SCALE}_{INTERMOD\ TONES}} \Rightarrow C/I = 27\,dB\right)$

c2) SNIR = $\frac{P_{TWO\ TONES}}{P_{THERM} + P_{INTERMOD}}$ AT OUTPUT of LNA? if $T_{sys} = 120°C = 393.15K$, $B = 250\,MHz$

$P^{LNA\ OUT}_{THERM} = K\,T_{sys}\cdot B\cdot G_{FLT}\cdot G_{LNA}$

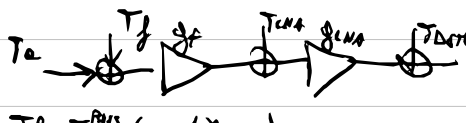
$1.356\,pW = -88.7\,dBm$ @ANTENNA PORT

$P^{LNA\ OUT}_{THERM} = -88.7\,dBm - 1\,dB + 28\,dB = -61.7\,dBm$

$$SNIR = \frac{10^{\frac{-50.5+3}{10}}\,mW}{10^{\frac{-77.5+3}{10}}\,mW + 10^{\frac{-61.7}{10}}\,mW} = \frac{1.778\cdot10^{-5}\,mW}{\underbrace{3.548\cdot10^{-8}\,mW + 6.76\cdot10^{-7}\,mW}_{0.7115\cdot10^{-6}\,mW}} =$$

$= 25 = 14\,dB$

d) Filter cooled @ -100°C ⇒ $T^{PHYS}_{FLT} = 173.15K$; $T_a = 30K$



$T_f = T^{PHYS}\left(\frac{1}{g_F} - 1\right) = 44.9K$ $\quad T_{sys} = T_a + T_f + \frac{T_{LNA}}{g_F} + \frac{T_{DOM}}{g_F g_{LNA}} = 393.15K$

$g_F = 10^{\frac{-f_F}{10}} = 0.7\%$ $\quad g_{LNA} = 10^{\frac{28}{10}} = 631$

$$\frac{T_{LNA}}{g_F} = T_{sys} - T_a - T_f - \frac{T_{DOM}}{g_F g_{LNA}} = 393.15K - 30 - 44.9K - 11K = 308K$$

$\Rightarrow T_{LNA} = 308K\cdot g_F = 244.5K$

$\Rightarrow NF_{LNA} = 1 + \frac{T_{LNA}}{T_0} = 1.84 \Rightarrow NF = 2.65\,dB$

NF of LNA is always defined as $NF = 1 + \frac{T_{LNA}}{290K}$ (h 16:43)

even if the LNA is COOLED